Trigonometric Functions & the Unit Circle

ANSWER KEY

Section 1: Radian Measure & Unit Circle Coordinates

1 What are the exact coordinates of the terminal point on the unit circle for angle $\theta = \frac{5\pi}{6}$?

a. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ ✓

b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

Angle $\frac{5\pi}{6}$ lies in Quadrant II with reference angle $\frac{\pi}{6}$. Therefore, $\cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$ and $\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$, giving coordinates $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$.

2 Convert an angle of $\frac{7\pi}{4}$ radians into degree measure.

a. 315° ✓

b. 225°

c. 300°

d. 135°

Multiply by $\frac{180^\circ}{\pi}$: $\frac{7\pi}{4} \times \frac{180^\circ}{\pi} = 7 \times 45^\circ = 315^\circ$.

3 If the terminal ray of angle α passes through the point $(-3, -4)$, what is $\sin\alpha$?

a. $-\frac{4}{5}$ ✓

b. $-\frac{3}{5}$

c. $\frac{4}{3}$

d. $\frac{3}{5}$

Radius $r = \sqrt{(-3)^2 + (-4)^2} = \sqrt{9 + 16} = 5$. Then $\sin\alpha = \frac{y}{r} = -\frac{4}{5}$.

Section 2: Exact Values of Trigonometric Functions

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Evaluate $\tan\left(\frac{4\pi}{3}\right)$.

a. $\sqrt{3}$ ✓

b. $-\sqrt{3}$

c. $\frac{\sqrt{3}}{3}$

d. $-\frac{\sqrt{3}}{3}$

Angle $\frac{4\pi}{3}$ is in Quadrant III, where tangent is positive. The reference angle is $\frac{\pi}{3}$. $\tan\left(\frac{\pi}{3}\right) = \sqrt{3}$, so $\tan\left(\frac{4\pi}{3}\right) = \sqrt{3}$.

5 What is the exact value of $\sec\left(-\frac{\pi}{4}\right)$?

a. $\sqrt{2}$ ✓

b. $-\sqrt{2}$

c. $\frac{\sqrt{2}}{2}$

d. 2

Cosine is an even function: $\cos(-\pi/4) = \cos(\pi/4) = \frac{\sqrt{2}}{2}$. Thus $\sec(-\pi/4) = \frac{1}{\cos(\pi/4)} = \frac{2}{\sqrt{2}} = \sqrt{2}$.

6 If $\sin\theta = \frac{3}{5}$ and $\cos\theta < 0$, find $\cot\theta$.

a. $-\frac{4}{3}$ ✓

b. $\frac{4}{3}$

c. $-\frac{3}{4}$

d. $-\frac{4}{5}$

Since $\sin\theta > 0$ and $\cos\theta < 0$, θ is in Quadrant II. Here $x = -\sqrt{5^2 - 3^2} = -4$. Then $\cot\theta = \frac{x}{y} = -\frac{4}{3}$.

Section 3: Sinusoidal Graphs & Transformations

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Find the amplitude and period of the sinusoidal function $f(x) = -4\cos(3x - \pi) + 2$.

a. **Amplitude** $= 4$**, Period** $= \frac{2\pi}{3}$ ✓

b. Amplitude $= -4$, Period $= 3$

c. Amplitude $= 4$, Period $= 3\pi$

d. Amplitude $= 2$, Period $= \frac{2\pi}{3}$

Amplitude is $|A| = |-4| = 4$. For $B = 3$, the period is $\frac{2\pi}{B} = \frac{2\pi}{3}$.

8 A Ferris wheel of radius 25 m completes one rotation every 60 seconds. Passengers board at the bottom, 2 m above the ground. Which cosine equation models height $h(t)$ in meters?

a. $h(t) = -25\cos\left(\frac{\pi}{30}t\right) + 27$ ✓

b. $h(t) = 25\cos\left(\frac{\pi}{30}t\right) + 27$

c. $h(t) = -25\cos\left(\frac{\pi}{60}t\right) + 2$

d. $h(t) = -27\cos\left(\frac{\pi}{30}t\right) + 25$

Centerline is $d = 2 + 25 = 27$. Amplitude is 25. Period is $60 \Rightarrow B = \frac{2\pi}{60} = \frac{\pi}{30}$. Since it starts at minimum height 2, an inverted cosine is used: $h(t) = -25\cos\left(\frac{\pi}{30}t\right) + 27$.

9 What is the phase shift of $g(x) = 2\sin\left(4x + \frac{\pi}{2}\right)$?

a. $\frac{\pi}{8}$ **units to the left** ✓

b. $\frac{\pi}{8}$ units to the right

c. $\frac{\pi}{2}$ units to the left

d. $\frac{\pi}{4}$ units to the left

Factor out 4: $g(x) = 2\sin\left(4\left(x + \frac{\pi}{8}\right)\right)$. The phase shift is $\frac{\pi}{8}$ units to the left.

Section 4: Fundamental Trigonometric Identities

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Simplify the trigonometric expression $\frac{\tan x}{\sec x}$.

a. $\sin x$ ✓

b. $\cos x$

c. $\csc x$

d. $\cot x$

Rewrite in terms of sine and cosine: $\frac{\sin x/\cos x}{1/\cos x} = \frac{\sin x}{\cos x} \cdot \cos x = \sin x$.

11 Which expression is equivalent to $(1 - \cos^2\theta)(1 + \cot^2\theta)$ for all defined θ?

a. 1 ✓

b. $\sin^2\theta$

c. $\tan^2\theta$

d. $\cos^2\theta$

Use Pythagorean identities: $1 - \cos^2\theta = \sin^2\theta$ and $1 + \cot^2\theta = \csc^2\theta$. Therefore, $\sin^2\theta \cdot \csc^2\theta = \sin^2\theta \cdot \frac{1}{\sin^2\theta} = 1$.

12 Solve for $\theta \in [0, 2\pi)$: $2\cos^2\theta - \cos\theta - 1 = 0$.

a. $\theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$ ✓

b. $\theta = \pi, \frac{\pi}{3}, \frac{5\pi}{3}$

c. $\theta = 0, \frac{\pi}{3}, \frac{5\pi}{3}$

d. $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$

Factor: $(2\cos\theta + 1)(\cos\theta - 1) = 0$. So $\cos\theta = 1 \Rightarrow \theta = 0$, and $\cos\theta = -\frac{1}{2} \Rightarrow \theta = \frac{2\pi}{3}, \frac{4\pi}{3}$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647